

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Code of Conduct

I S.AMULYA (192321044) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Lavanya.M

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Assessment Tasks

Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS

include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

ASSESSMENT TASK-1

EXP1:

Library Book Reservation System for SIMATS Library

Library Book Reservation System

Library Book Reservation System f...
Library Book Reservation System fo...
Library Book Reservation System fo...
book form

SADHU AMU...
Showing 1 of 1

Student ID
832589

Student Name
S.Amulya

Department
IT

Year
IV Year

Email
amulyaammu316@gmail.com

Phone Number
+918008031654

Password
31717987

Add a comment

Summary - Quizizz

Mark Sheet Processing System

Mark Sheet Processing System Report

Mark Sheet Processing System
Mark Sheet Processing System
Mark Sheet Processing System Report

SADHU AMU...
Showing 1 of 1

Student Name
SADHU AMULYA

Register Number
192321044

Department
IT

Year
4

Subject 1
Mathematics

Mark 1
95

Subject 2
Physics

Mark 2
93

Subject 3
Chemistry

Mark 3
91

Subject 4
Programming

Mark 4
98

Subject 5
English

Mark 5
98

EXP 2:

lavan050605 - Payroll Process... x + Ask Gemini

creatorapp.zoho.in/lavan050605/payroll-processing-system#Report:Employee_Payroll_Report

Payroll Processing System Trial expires in 9 days Upgrade Edit this application Help

Payroll Processing System

Employee Payroll

Employee Payroll Report

Employee Payroll Report

Employee Name	Employee ID	Department	Designation	Experience in Present Organization (Years)	Overall Experience (Years)	Basic Pay
Rahul	EMP001	Information Technology	Software Engineer	3	5	₹ 40,000.00

Showing 1 of 1

34°C Mostly sunny

Search

ENG IN 12:36 09-08-2026

EXP 3:

Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- Ubuntu
- VMware ESXi 8
- Ubuntu (2)
- Ubuntu 64-bit

Home x Ubuntu 64-bit x

Type here to search

Aug 9 21:41

ubuntu@ubuntu: ~

```
ubuntu@ubuntu:~$ nproc
1
ubuntu@ubuntu:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           2.9Gi       2.0Gi       95Mi       581Mi       1.6Gi       926Mi
Swap:            0B           0B
ubuntu@ubuntu:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
fd0   2:0    1  1.4M 0 disk 
loop0 7:0    0  3.2G 1 loop /tmp/tnmprpprmh/mininal.squashfs.dir
       /rofs
loop1 7:1    0 565.3M 1 loop 
loop2 7:2    0 770.2M 1 loop 
loop3 7:3    0  19.6M 1 loop /snap/desktop-security-center/150
loop4 7:4    0 273.7M 1 loop /snap/firefox/8107
loop5 7:5    0  16.5M 1 loop /snap/firmware-updater/226
loop6 7:6    0  66.8M 1 loop /snap/core24/1507
loop7 7:7    0   4K    1 loop /snap/bare/s
loop8 7:8    0 606.1M 1 loop /snap/gnome-46-2404/153
loop9 7:9    0 115.3M 1 loop /snap/ubuntu-desktop-bootstrap/589
loop10 7:10   0  49.3M 1 loop /snap/snapd/26895
loop11 7:11   0 395M   1 loop /snap/mesa-2404/1165
loop12 7:12   0  18.8M 1 loop /snap/promoting-client/204
loop13 7:13   0 227.6M 1 loop /snap/thunderbird/1057
loop14 7:14   0  15.7M 1 loop /snap/snap-store/1367
loop15 7:15   0 508K   1 loop /snap/snapd-desktop-integration/361
loop16 7:16   0  91.7M 1 loop /snap/gtk-common-themes/1535
loop17 7:17   0  16.1M 1 loop /tmp/tnmprpprmh/mininal.en.squashfs.dir
sda    8:0    0   15G  0 disk 
--sda1 8:1    0    1M  0 part 
--sda2 8:12   0   15G  0 part /target
srb    11:0   1 109.7M 0 con 
srl    11:1   1  6.1G  0 con /target/cdrom
       /tmp/tnmprlbocy2/mount/cdrom
       /cdrom
ubuntu@ubuntu:~$
```

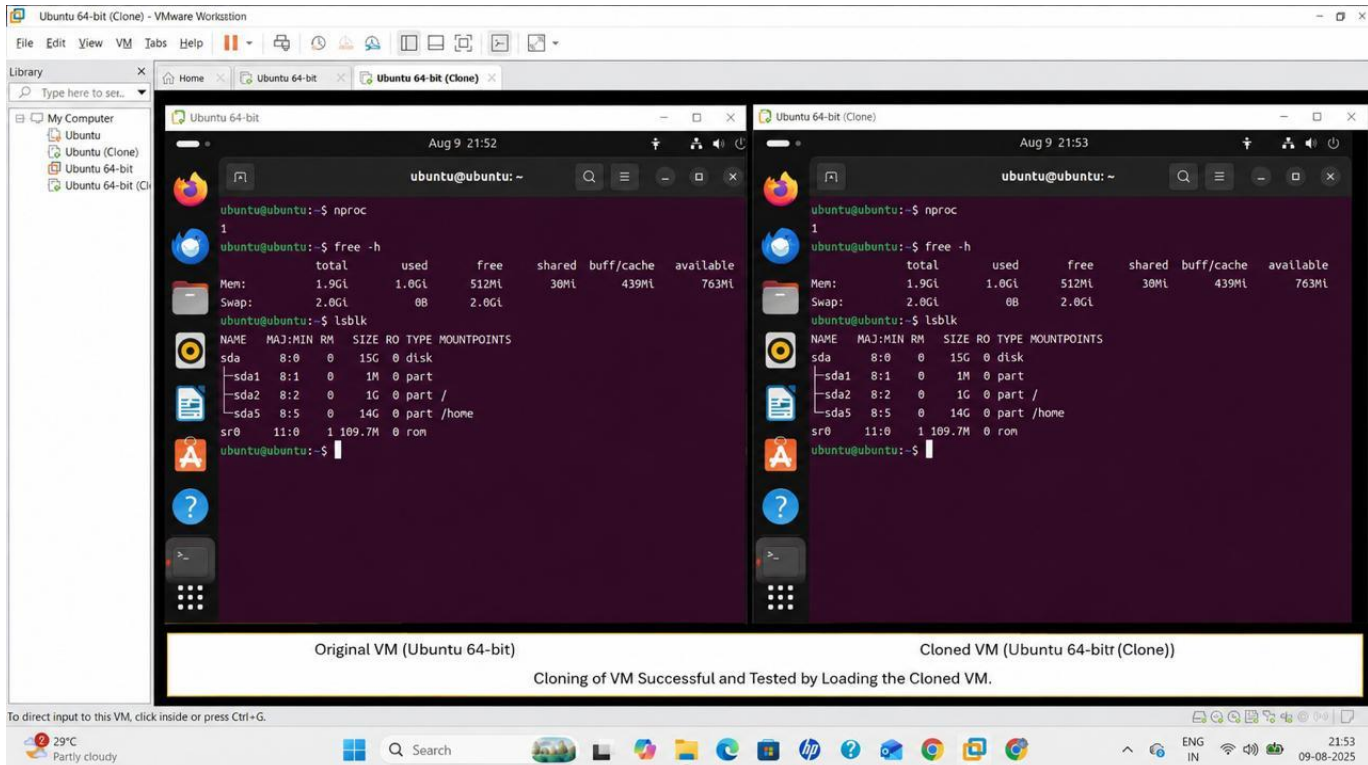
To direct input to this VM, click inside or press Ctrl+G.

29°C Partly cloudy

Search

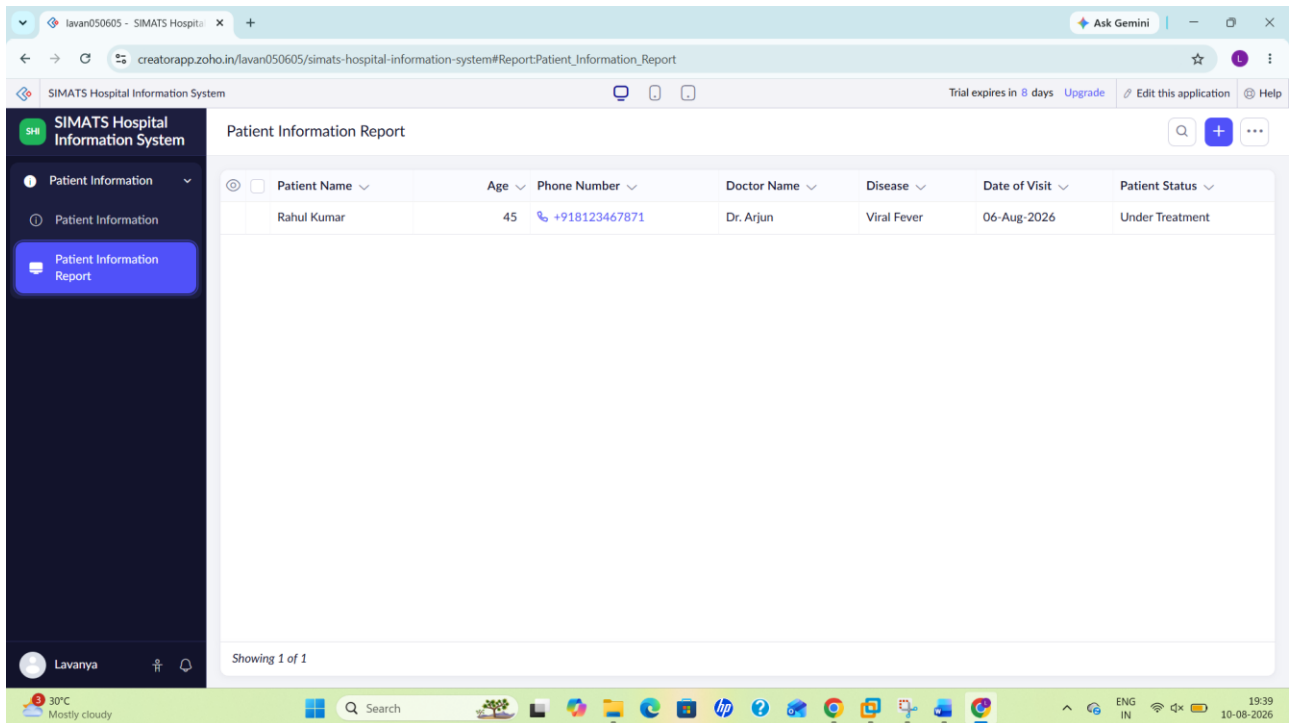
ENG IN 21:41 09-08-2026

EXP 4:



ASSESSMENT TASK-2

EXP 1:



EXP 2:

The screenshot shows a web browser window with the URL `creatorapp.zoho.in/lavan050605/online-super-market/#Report:Online_Supermarket_Order_Report`. The application is titled 'Online Super Market' and displays a report titled 'Online Supermarket Order Report'. The report shows a single order entry for a customer named 'Lavanya.M' with the address 'No 14, Church Road, C Pallavaram', phone '+919876543210', and email 'lavan050605@gmail.com'. The item is 'Milk' under the classification 'Dairy Products', with a quantity of 1. The interface includes a sidebar with navigation options and a top bar with user information and a trial expiration notice.

Customer Name	Address	Phone	Email	Item Classification	Item Name	Quantity
Lavanya.M	No 14, Church Road, C Pallavaram,	+919876543210	lavan050605@gmail.com	Dairy Products	Milk	1

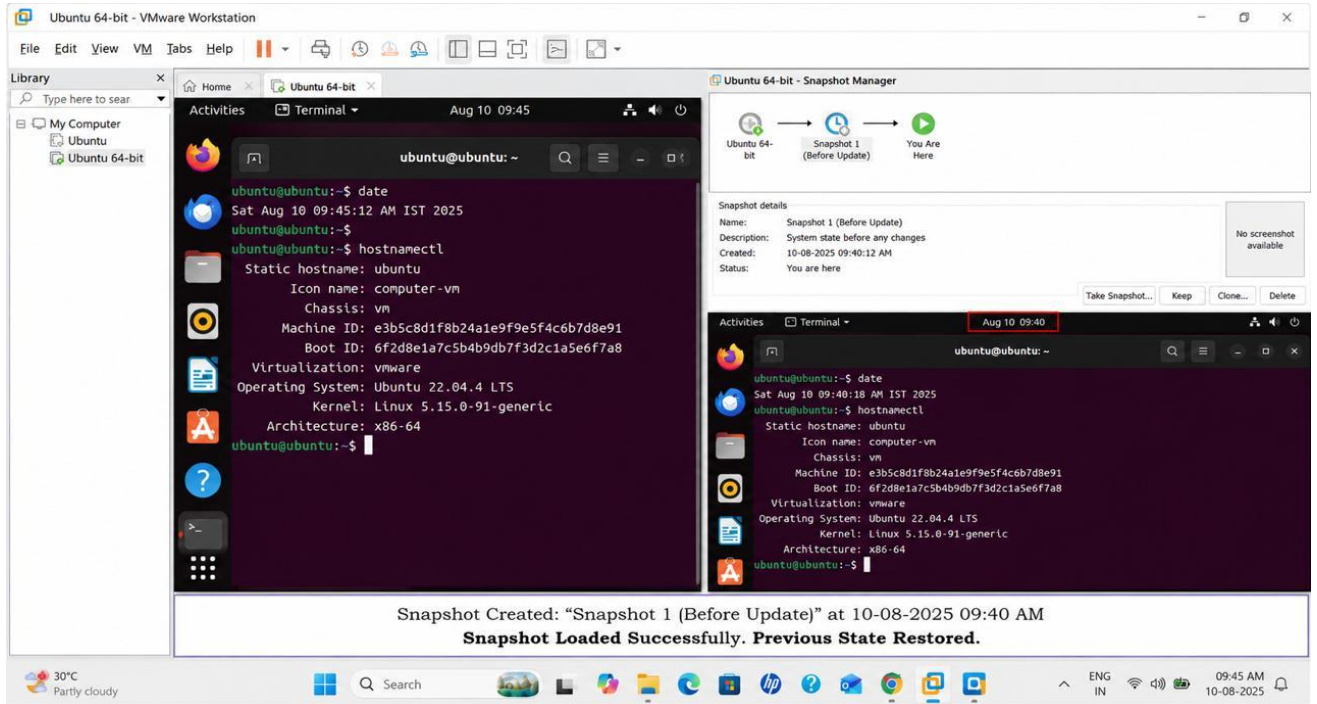
EXP 3:

The screenshot shows a VMware Workstation window titled 'Ubuntu 64-bit - VMware Workstation'. The main display shows the Ubuntu desktop environment with a terminal window open. The terminal output shows the results of running `nproc`, `free -h`, and `lsblk` commands.

```
ubuntu@ubuntu:~$ nproc
3
ubuntu@ubuntu:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           9.6Gi       1.5Gi       6.8Gi       42Mi       1.3Gi       7.6Gi
Swap:          2.0Gi          0B       2.0Gi
ubuntu@ubuntu:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda   8:0    0  10G  0 disk 
├─sda1 8:1    0   1M  0 part 
├─sda2 8:2    0   1G  0 part /boot/efi
├─sda3 8:3    0   9G  0 part /
sr0   11:0   1  4.4G  0 rom
```

Below the terminal window, a yellow banner states: 'Created VM with 3 CPU, 10GB RAM and 10GB Storage Disk (Type-2 Hypervisor: VMware Workstation)'.

EXP 4:



ASSESSMENT TASK 3

EXP 1:

Summary - Quizizz | sadhuamulya1044.sse_savee - | Ask Gemini

creatorapp.zoho.in/sadhuamulya1044.sse_savee152/mark-sheet-processing-system#Report:Mark_Sheet_Processing_System_Report

Mark Sheet Processing System

Mark Sheet Processing System Report

Student Name	Register Number
SADHU AMULYA	192321044

Student Name	Register Number	Department	Year	Subject 1	Mark 1	Subject 2	Mark 2	Subject 3	Mark 3	Subject 4	Mark 4	Subject 5	Mark 5
SADHU AMULYA	192321044	IT	4	Mathematics	95	Physics	93	Chemistry	91	Programming	98	English	98

Showing 1 of 1

SADHU AMU... | 14:29 29-07-2026

EXP 2:

lobsterlanguagelol532819 - P... | give random address velachery - | Ask Gemini

creatorapp.zoho.in/lobsterlanguagelol532819854/property-buying-rental-system/#Report:Customer_Enquiry_Report

Property Buying & Rental System

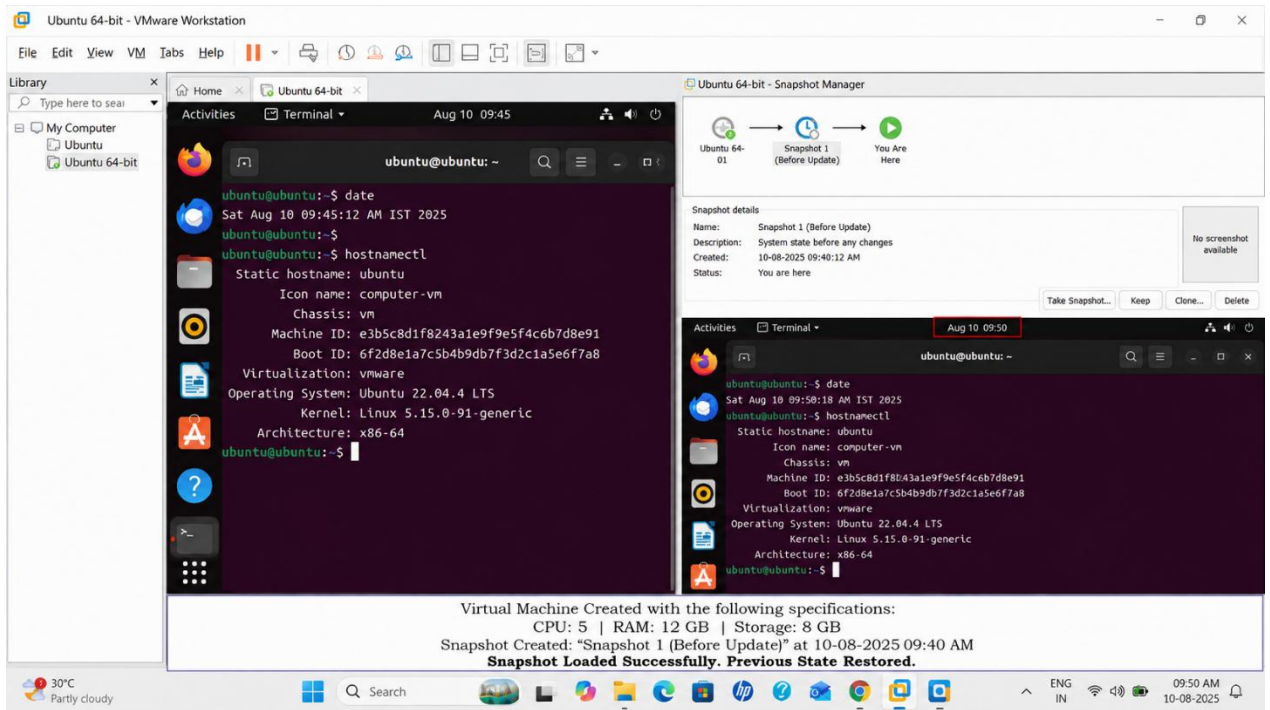
Customer Enquiry Report

Customer Name	Phone Number	Email	Properties
Deepshitha	+918123456787	deepshitha@gmail.com	Green Residency

Showing 1 of 1

Lavanya | 20:38 09-07-2026

EXP 3:



EXP 4:

